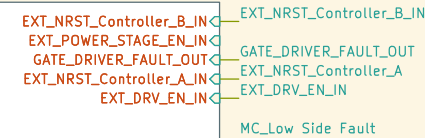


HyperDrive

Mk. 1

Isolated Driver Interface



File: Isolated_Driver_Interface.kicad_sch

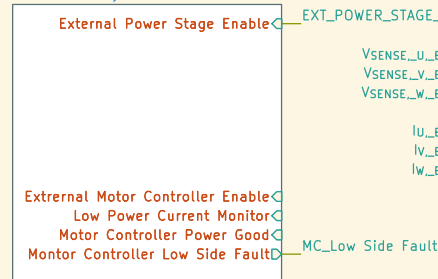
©TODO:
- Add fan header.
- Add smd spacer mounts and footprints

Communications



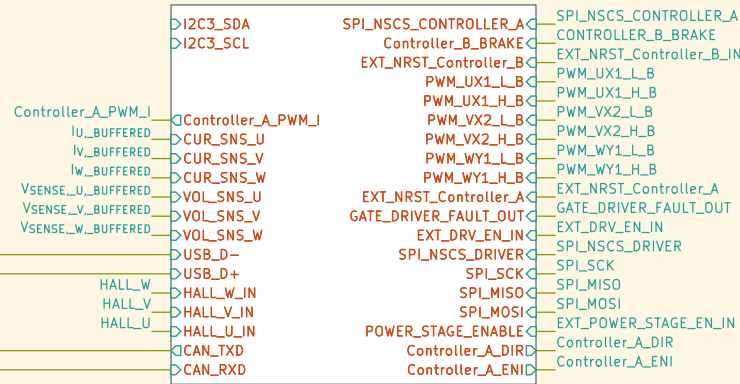
File: Communication.kicad_sch

Power Delivery



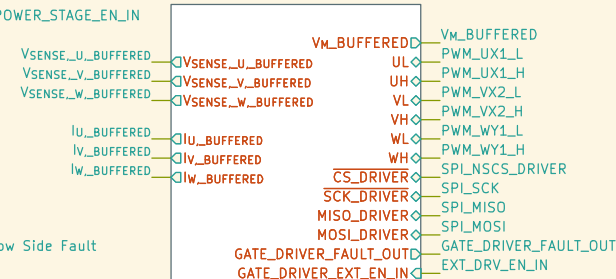
File: Power Delivery.kicad_sch

Microcontroller



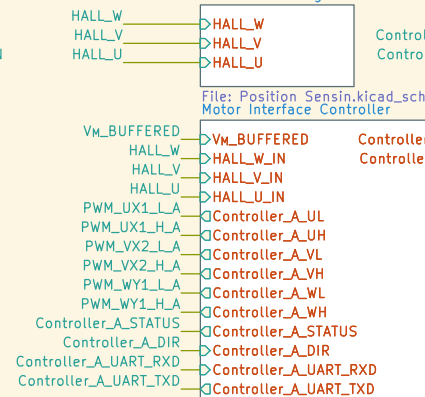
File: Microcontroller.kicad_sch

Motor Gate Driver



File: Motor Gate Driver and Power Stage.kicad_sch

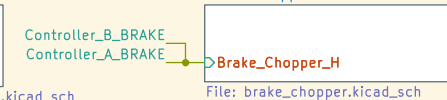
Position Sensing



File: Position Sensin.kicad_sch
Motor Interface Controller

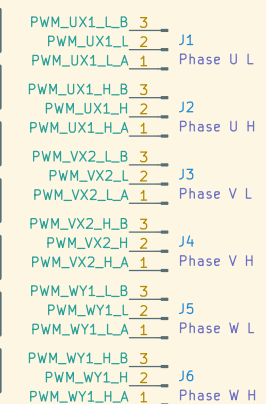
Controller_A_COIL_U
Controller_A_COIL_V

Brake Chopper



File: brake_chopper.kicad_sch

Controller Interconnect



©Application Note
Digital Signal Jumpers must be pin headers to allow for on the fly EN/DIS

- H7 MountingHole
- H8 MountingHole
- H9 MountingHole
- H10 MountingHole



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Sheet: /
File: Motor Controller.kicad_sch

Title: HyperDrive Motor Controller

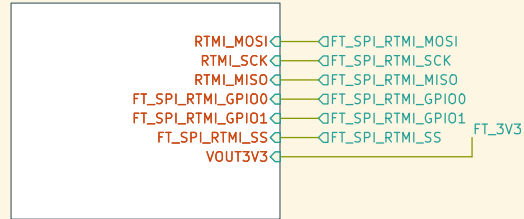
Size: A4 Date: 2024-05-30

KiCad E.D.A. 8.0.4

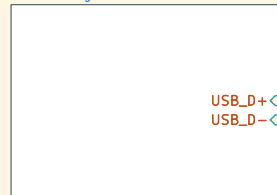
Rev: Mk. 1

Id: 1/17

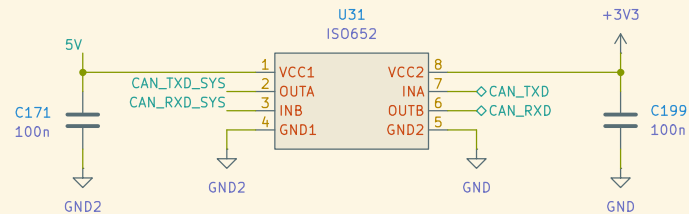
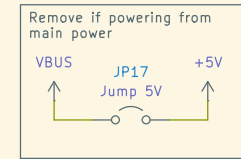
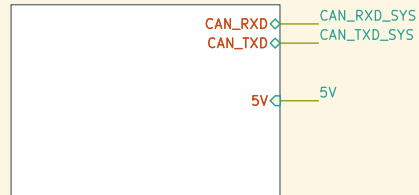
RTMI Interface



USB Debug Interface



CAN Bus Interface



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Sheet: /Communications/

File: Communication.kicad_sch

Title: HyperDrive Motor Controller

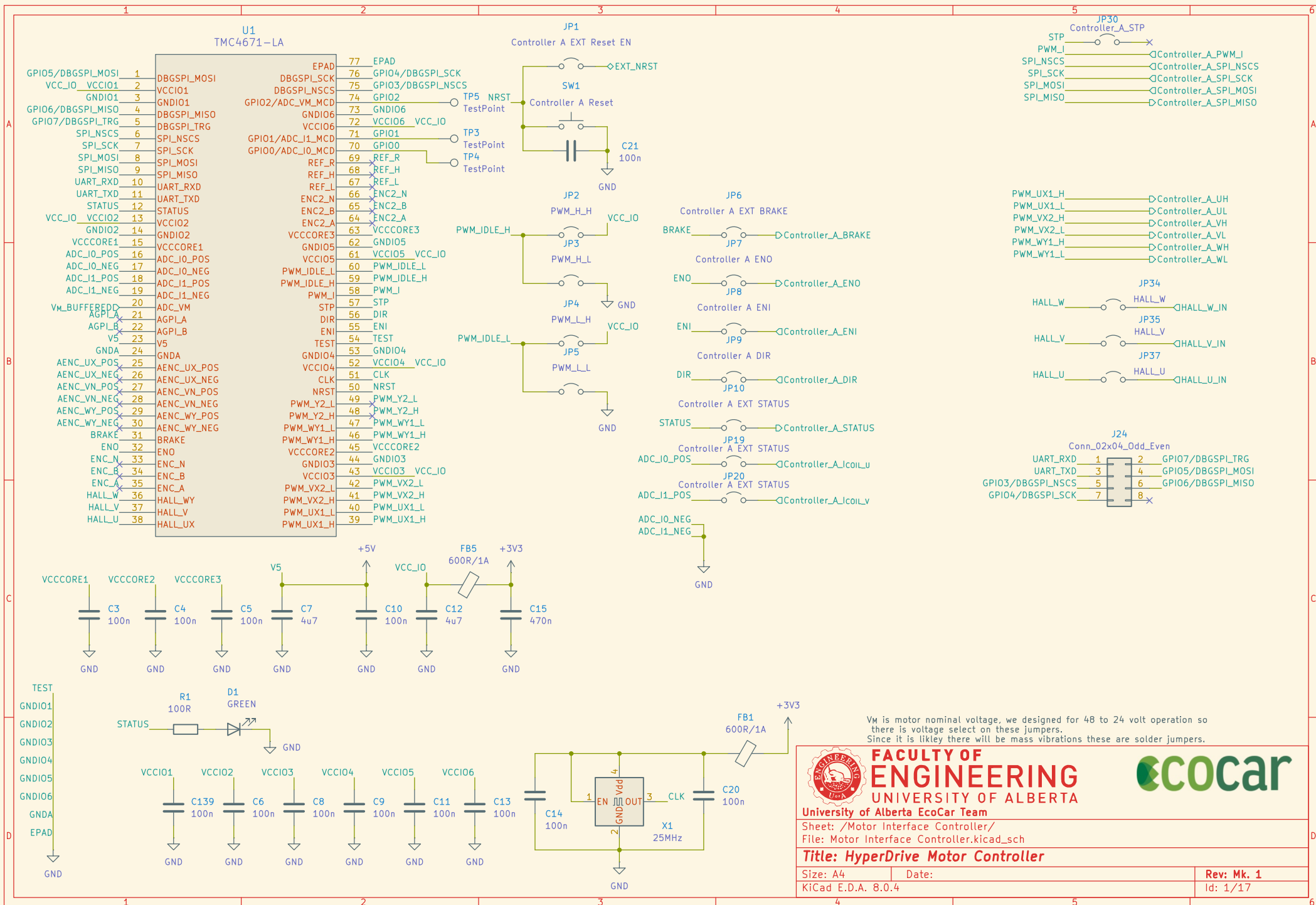
Size: A4

Date: 2024-05-16

Rev: **Mk. 1**

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Id: 1/17



Vm is motor nominal voltage, we designed for 48 to 24 volt operation so there is voltage select on these jumpers. Since it is likely there will be mass vibrations these are solder jumpers.



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Sheet: /Motor Interface Controller/
File: Motor Interface Controller.kicad_sch

Title: HyperDrive Motor Controller

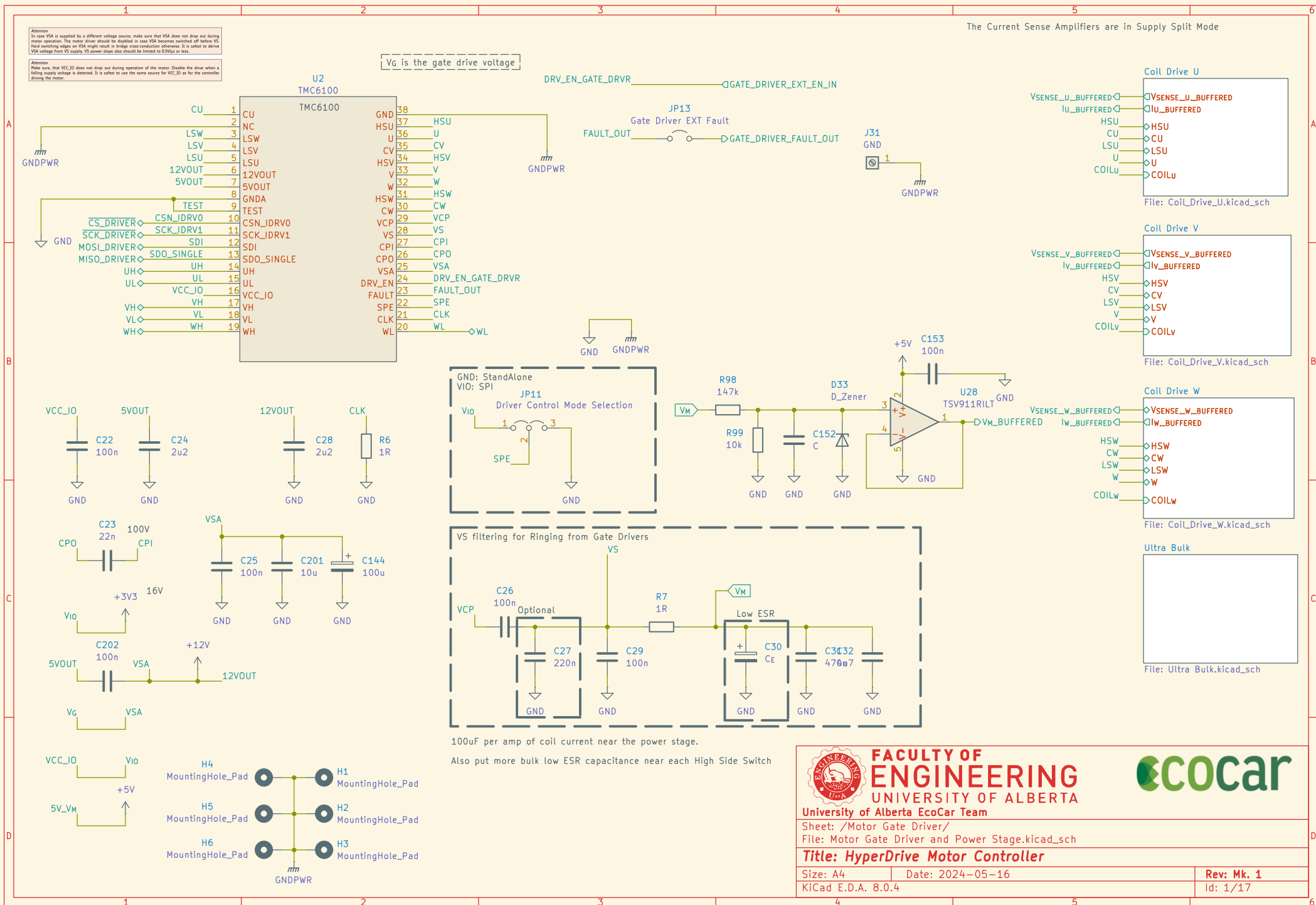
Size: A4

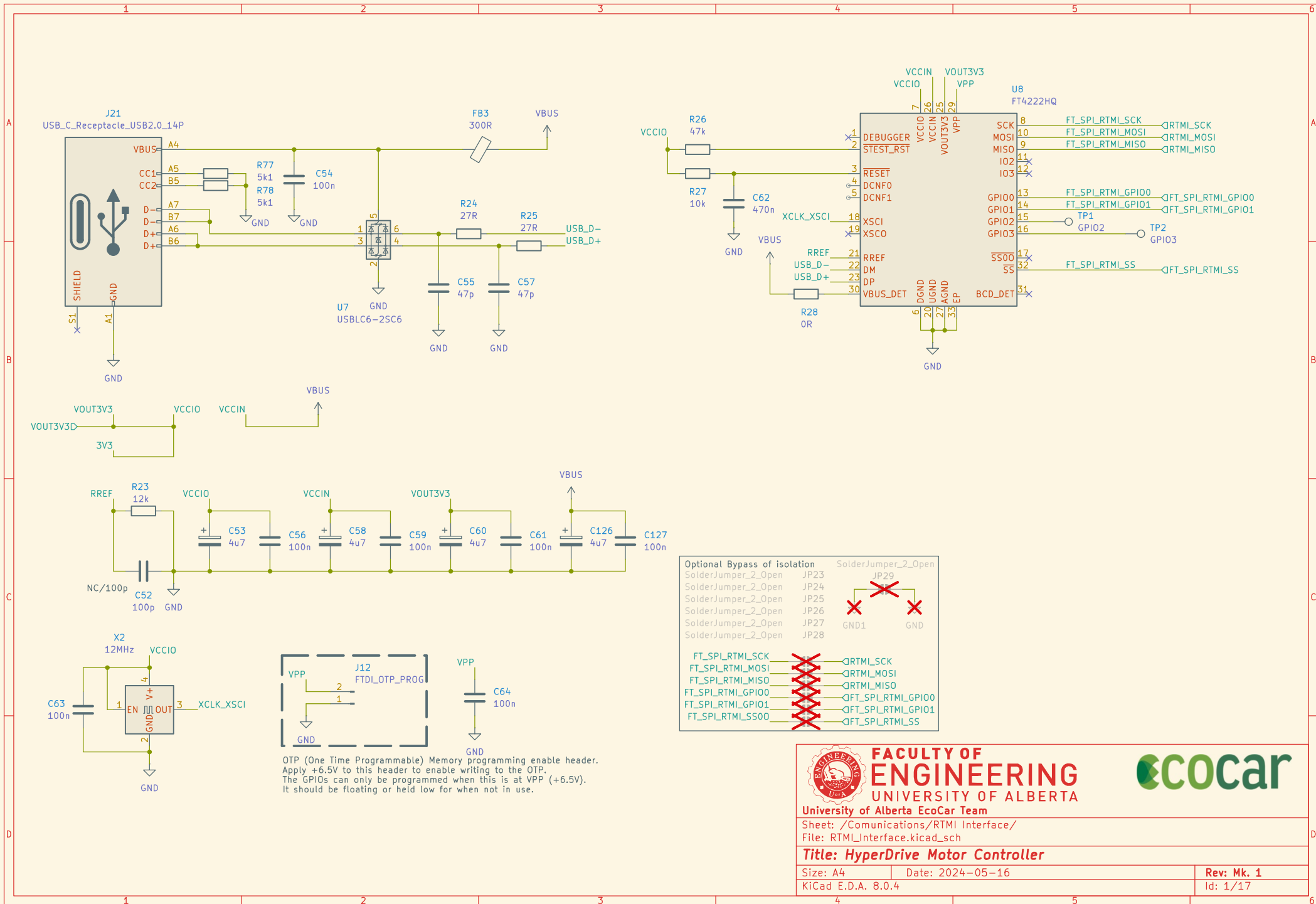
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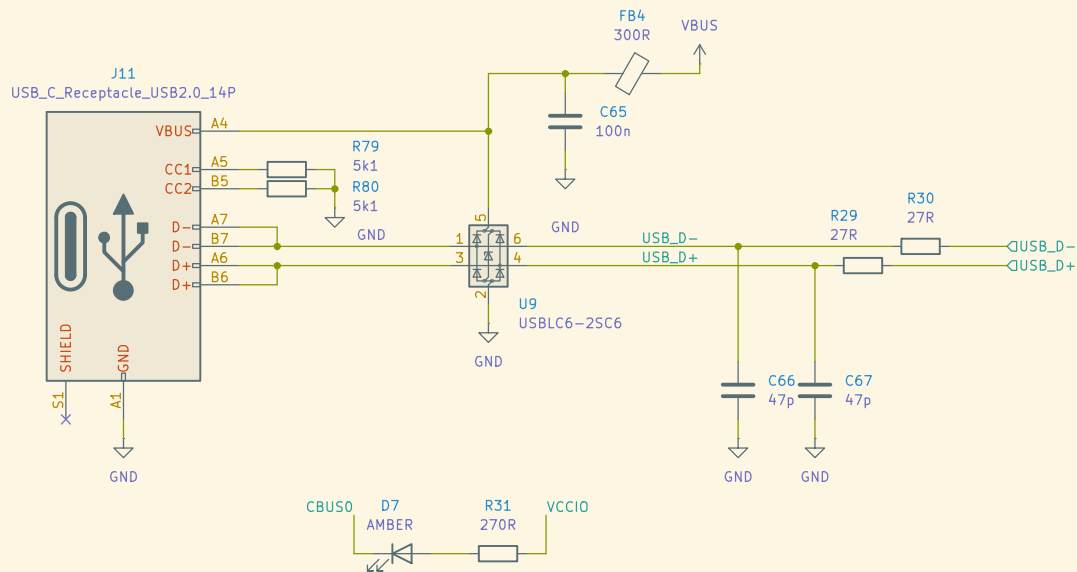
KiCad E.D.A. 8.0.4

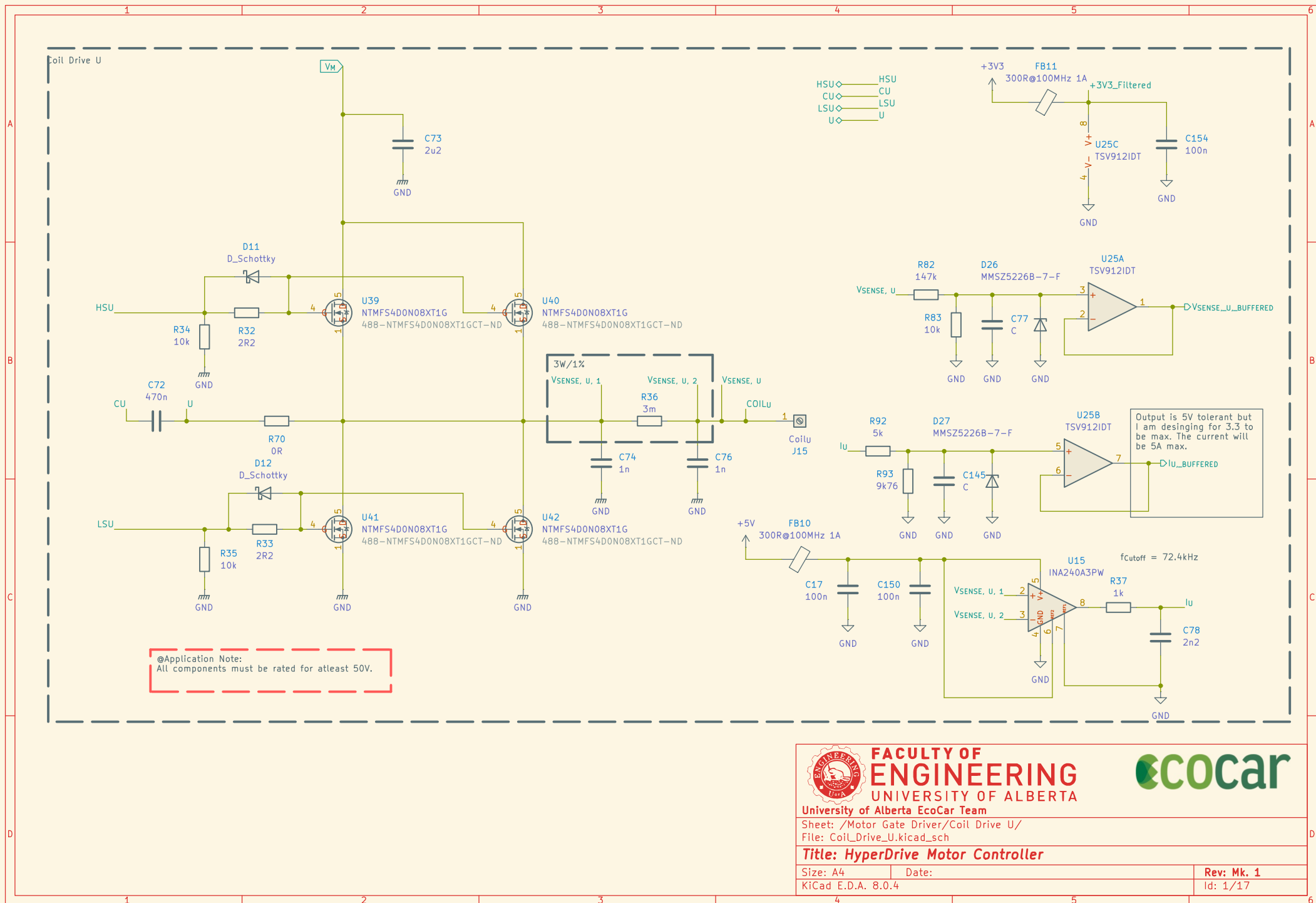
Rev: Mk. 1

Id: 1/17

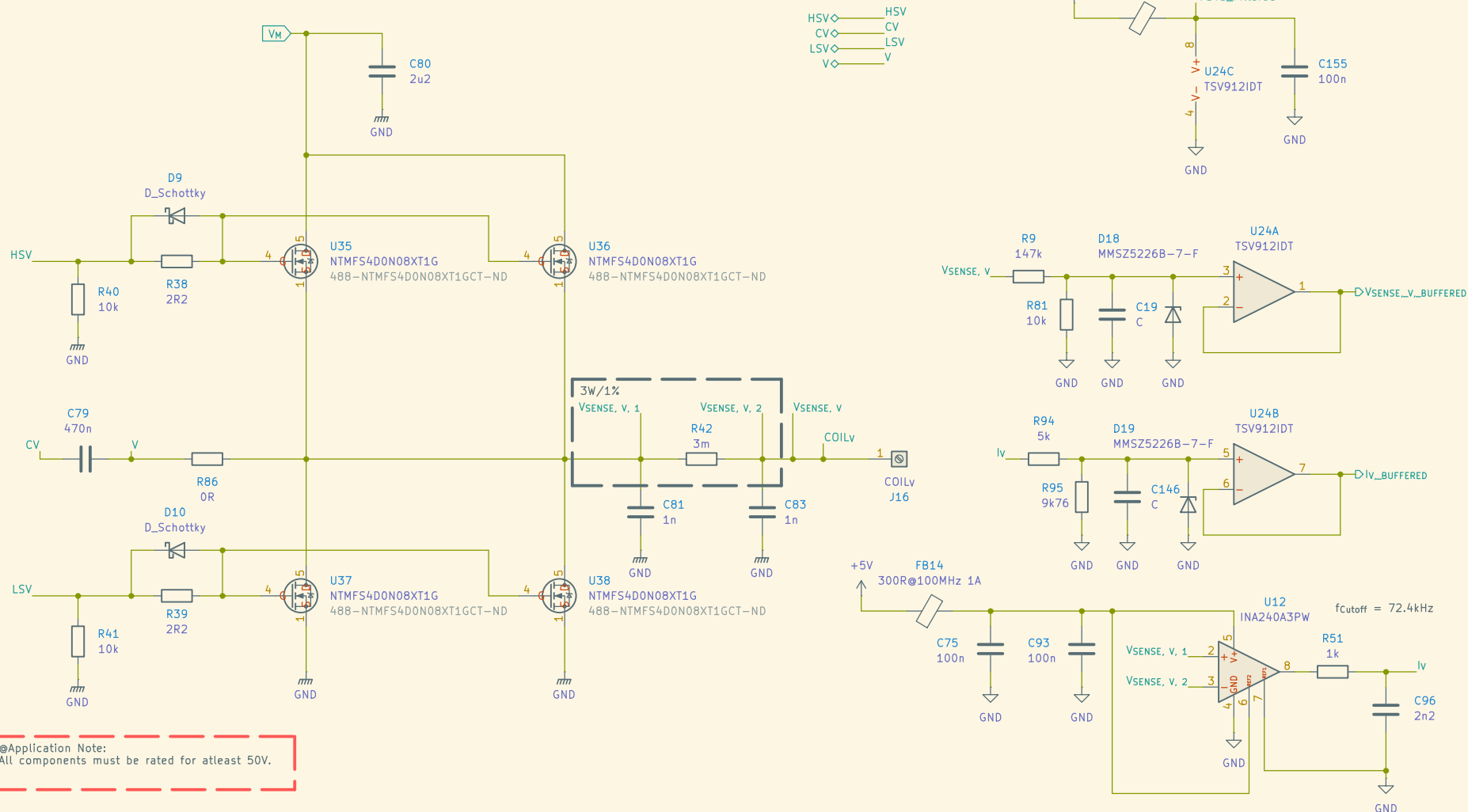








Coil Drive V



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Sheet: /Motor Gate Driver/Coil Drive V/

File: Coil_Drive_V.kicad_sch

Title: HyperDrive Motor Controller

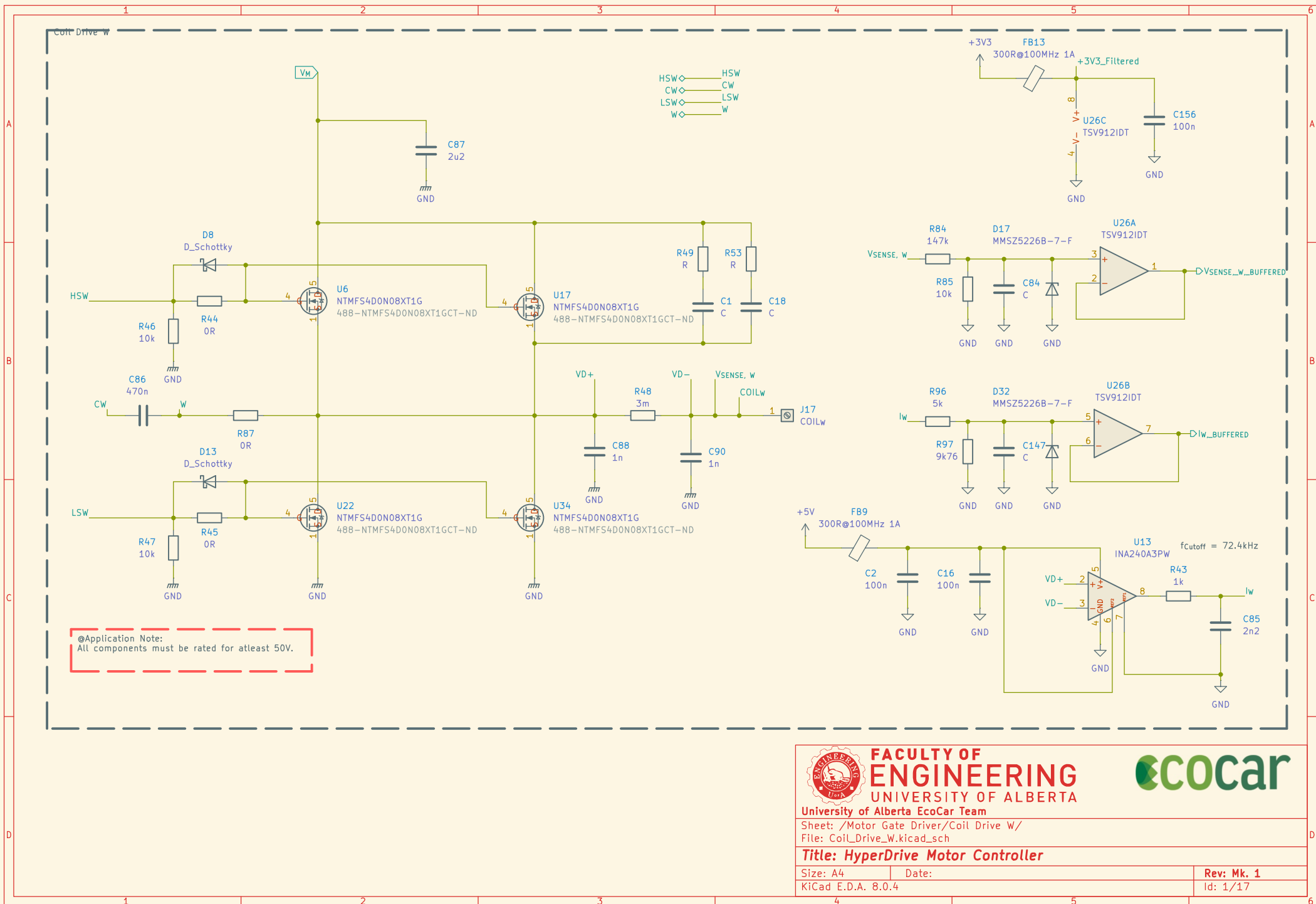
Size: A4

Date:

KiCad E.D.A. 8.0.4

Rev: Mk. 1

Id: 1/17



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Sheet: /Motor Gate Driver/Coil Drive W/

File: Coil_Drive_W.kicad_sch

Title: HyperDrive Motor Controller

Size: A4

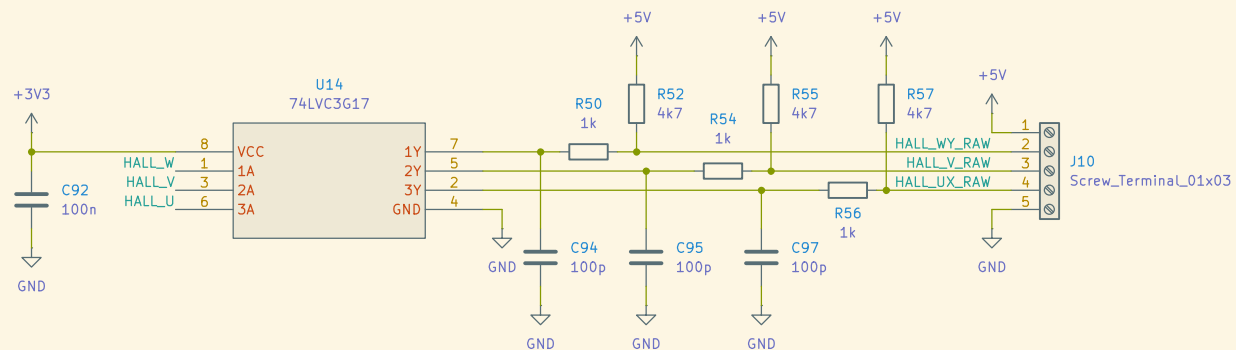
Date:

KiCad E.D.A. 8.0.4

Rev: Mk. 1

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HALL_W — CHALL_W
HALL_V — CHALL_V
HALL_U — CHALL_U



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Sheet: /Position Sensing/

File: Position Sensin.kicad_sch

Title: HyperDrive Motor Controller

Size: A4

Date:

Rev: **Mk. 1**

KiCad E.D.A. 8.0.4

Id: 1/17

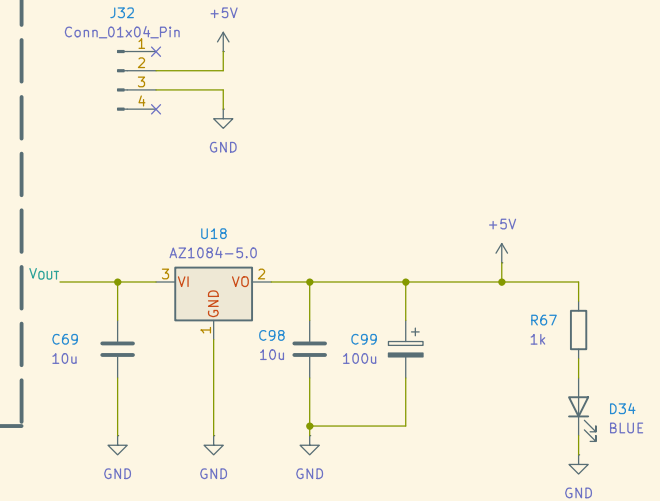
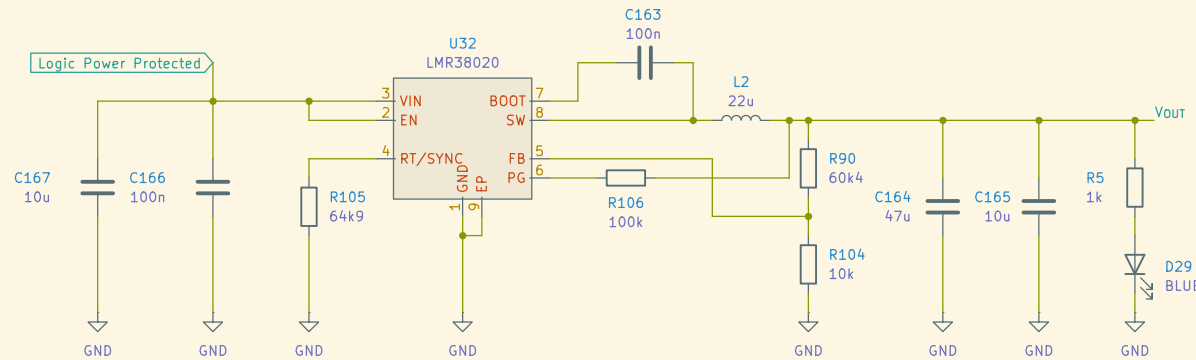


Power Regulation

This is a DC-DC Converter able to operate from either Motor supply or Battery supply.
It converts Input to 7V for use on the inverter controllers. Ultra Efficient.

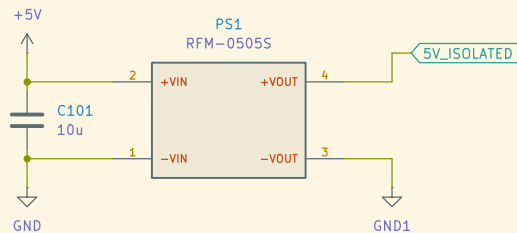
LMR38020 Buck Regulator

Compact DC-DC Converter 7V Vout



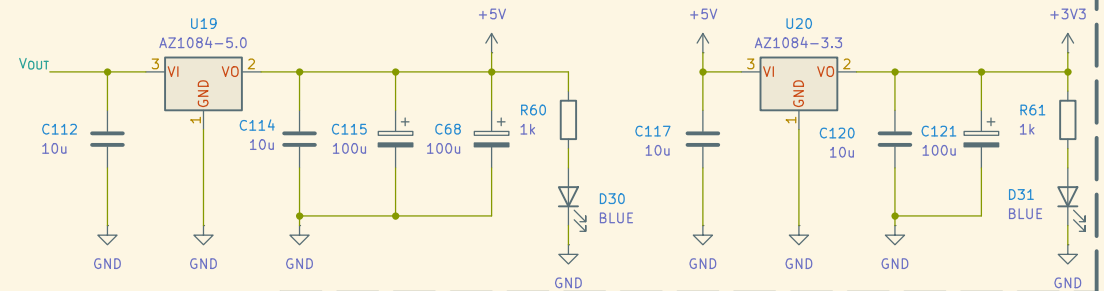
RFM-0505S Isolated DC-DC

Compact Isolated DC-DC Module



AZ1084 Linear Regulator

Low Drop Out Linear Regulators 5A @ 1.35VdROPOUT



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Sheet: /Power Delivery/Power Regulators/

File: power_regulators.kicad_sch

Title: HyperDrive Motor Controller

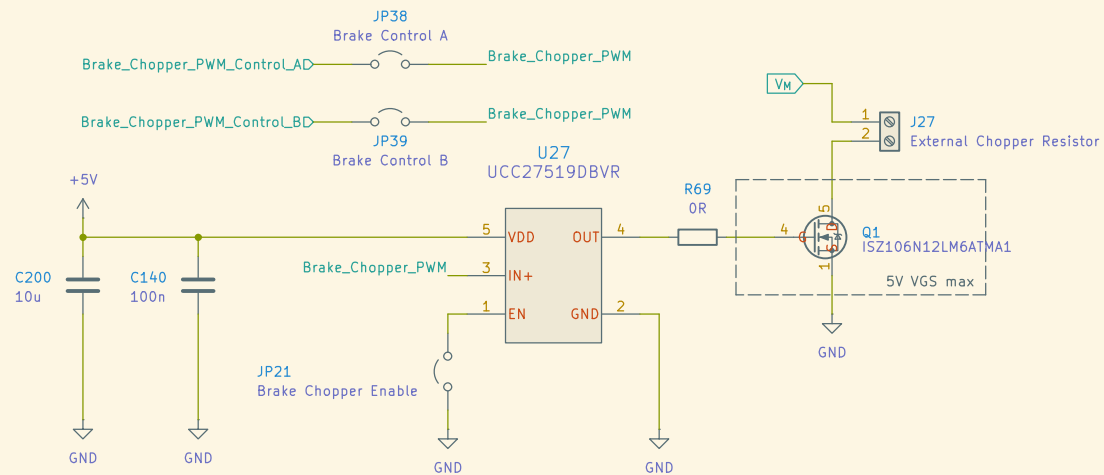
Size: A4

Date:

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Rev: Mk. 1

Id: 1/17



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Sheet: /Brake Chopper/

File: brake_chopper.kicad_sch

Title: HyperDrive Motor Controller

Size: A4

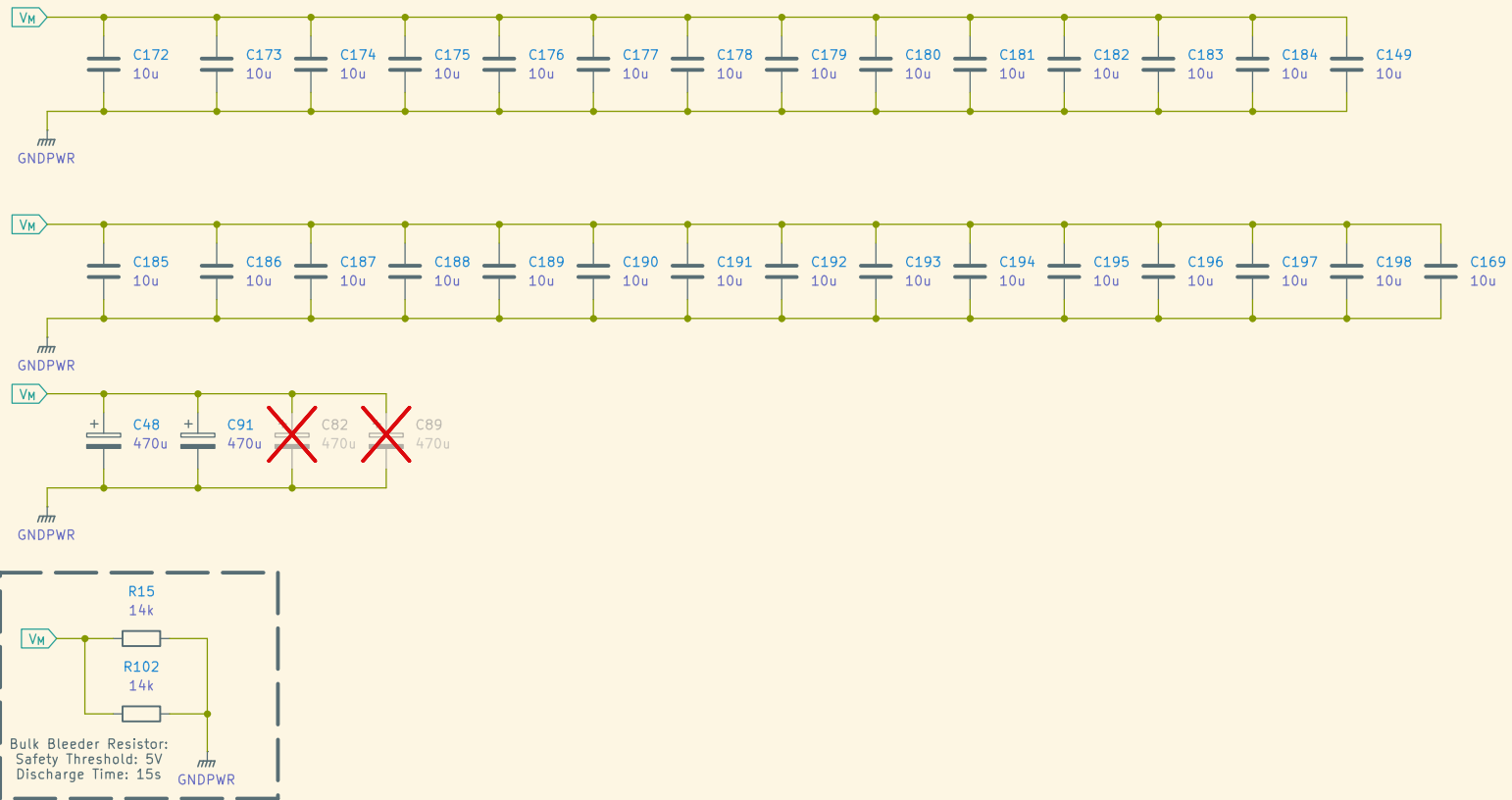
Date:

KiCad E.D.A. 8.0.4

Rev: **Mk. 1**

Id: 1/17





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Sheet: /Motor Gate Driver/Ultra Bulk/

File: Ultra Bulk.kicad_sch

Title: HyperDrive Motor Controller

Size: A4

Date:

Rev: Mk. 1

KiCad E.D.A. 8.0.4

Id: 1/17

